Dual Space 对偶空间

History regular polyhedron

$\begin{cases}\text{正六面体} \\ \text{正八面体}\end{cases}$ 对偶

正四面体 self-dual

$\begin{cases}\text{正十二面体} \\ \text{正二十面体}\end{cases}$

$n=4 \longrightarrow 6$ 个

$n \geq 5 \longrightarrow 3$ 个

$V/\mathbb{F}$ $\mathrm{Hom}(V,\mathbb{F}) = V^* \longrightarrow \mathrm{Hom}(V,\mathbb{F}^n) \cong (V^*)^{\times n}$

$\dim V = n < \infty$, choose a basis $v_1, v_2, \cdots v_n$ of V

Define $v_i^* \in V^*$ by $(v_i^*)v_j = \delta_{ij} \Rightarrow v_1^*, \cdots v_n^*$ is a basis of V^*

$\forall \sigma \in V^*$, then $\sigma = \sum_{i=1}^{n} x_i v_i^*$ $x_i = \sigma(v_i)$

· $\forall S \in \mathcal{L}(V)$ $S^\circ = \{\sigma \in V^* \mid \sigma(s) = 0, \forall s \in S\}$ annihilator

$\forall U \in \mathcal{L}(V^*)$ $\overset{\text{ann}}{\longleftarrow}$ $\ker U = \bigcap_{\sigma \in U} \ker \sigma$

Galois connection (伽罗华联络) $S_1 \subseteq S_2 \Rightarrow S_1^\circ \supseteq S_2^\circ$

$U_1 \subseteq U_2 \Rightarrow \ker U_1 \supseteq \ker U_2$

	$\mathcal{L}(V)$		$\mathcal{L}(V^*)$
n	$V = V_n$	n	$V = V_n^*$
$n-1$	$V_{n-1} \cdots$	$n-1$	U_{n-1}^*
$n-2$	$V_{n-2} \cdots$	$n-2$	U_{n-2}^*
$\vdots$		$\vdots$	
2	V_2	2	U_2^*
1	V_1	1	V_1^*
0	0		0

ann

ker

Date / /

$$V \xrightarrow[A_{m\times n}]{\sigma} W$$

$$V \xleftarrow[A^T_{n\times m}]{\sigma^*} W$$

dual of σ induced by σ

$$\begin{tikzcd} V \arrow[r, "f\circ\sigma = \sigma^*(f)"] \arrow[d, "\sigma"'] & F \\ W \arrow[r, "f"] & F \end{tikzcd}$$

Thm Let $\sigma \in Hom(V, W)$. Let A be the matrix of σ under bases β_V and β_W

Then the dual map σ^* of σ has matrix A^T under the dual bases β_W^* and β_V^*

General: $Hom(-, X)$ $\quad V \xrightarrow{\sigma} W$

$$Hom(V, X) \xleftarrow{\sigma^* = Hom(\sigma, X)} Hom(W, X)$$

$$\sigma^*(f) = f\circ\sigma \longleftarrow f$$

- $Hom(X, -)$ $\quad V \xrightarrow{\sigma} W$

$$Hom(X, V) \xrightarrow{\sigma_* = Hom(X,\sigma)} Hom(X, W)$$

$$f \longmapsto \sigma\circ f = \sigma_*(f)$$

CH V. Eigen-Theory of Linear operations

EX ① $\rho = \frac{d}{dx} : C^\infty \longrightarrow C^\infty$

$f(x) \longmapsto f'(x)$

$f'(x) = \rho(f(x)) = \lambda f(x) \Rightarrow f(x) = ce^{\lambda x}$

so $\sigma(\rho) = \mathbb{R}$

② $\rho = \frac{d}{dx} : \mathbb{R}[x]_n \longrightarrow \mathbb{R}[x]_n$

$\sigma(\rho) = \{0\}$

conclusion Let $V/\mathbb{F}$ be f.d. Then $\forall f \in \mathrm{End}\,V$ can be defined by $\sigma(A)$

where A is the matrix of σ under any basis of V

Pf. Let $v_1, \dots v_n$ be a basis of V

Let A be the matrix of f under this basis

i.e. $f(v_1, v_2 \cdots v_n) = (v_1, \dots v_n) A$

Suppose $\lambda \in \sigma(f)$ i.e. $\exists 0 \neq v$ s.t. $\sigma(v) = \lambda v$

$v = (v_1, \dots v_n) X$

$\sigma(v) = \sigma(v_1, v_2 \cdots v_n) X = (v_1, v_2 \cdots v_n) AX$

$\lambda v = (v_1, v_2 \cdots v_n) \lambda X$

$AX = vX \Leftrightarrow \sigma(v) = \lambda v$